

Advanced Macroeconomics

Lecture 0 : Review of Macroeconomics and Keynesian Economics

MSc/MA Economics

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Monday 21 September, 2026

Pre Keynesian Macroeconomics literature

Thomas Robert Malthus(1766-1834)

- Suggested the possibility of crisis of effective demand.
- In Classical economics Wages are determined by VALUE OF LABOR, i.e. Bare minimum which is require for reproduction of Labor Power and also in long run wages are governed by “Iron law of wages”, i.e. wages tend to converge at the level of bare minimum subsistence.
- Also with rising population leads to increase in demand for food, this implies that more and more less fertile land is used into the cultivation , therefore a diminishing supply and rising prices for food.
- with rising food price, the bare minimum subsistence level itself inc., which means a falling profit of Capitalist.
- Now with wages at bare minimum level and falling/ lower profit of capitalist.. who will demand the commoditiesMALTHUS solution is LANDLORDS.

- David Ricardo(1772-1823), a contemporary of Malthus provides solution in term of free trade, i.e. free trade of agricultural surplus across the nations.

Karl MARX(1818-1883)

- In Marxian economics there are two types of economic crisis. 1) Realization crisis, and 2) profitability Crisis.
- Realization crisis:
- Is a short run phenomena.
- Emerges because of disproportionality b/w the investment and consumption good sectors.
- Also many of the orthodox Marxist believe that because labor is paid less than the value they produces and this means that larger share of national income is with capitalist and because MPC for capitalist low this implies falling effective demand.

- Marxian solution to the problem of Realization is to set up the link b/w the two departments and for falling effective demand there should be increase in exogenous demand, e.g. Govt. expenditure.
- Profitability crisis:
- Is a long run phenomena.
- Marx(capital 1867), held the view that in long run , there is a tendency in capitalism of falling profit, because of rising organic composition of capital i.e. the ratio of constant to total capital and since in Marxism only variable capital produces surplus(unrealized profit) therefore, a rising share of constant capital implies a falling profit.
- Marx's believed that this phenomena is inevitable in capitalist development, and there is no solution to this problem except for some temporary fixation and this eventually lead to demise of capitalist system

Economic theory before Keynesian Revolution(1870-1936)

- The year 1871 marked a turning point in the history of economic thought. It witnessed the end of classical economics and rise of new school of economic thought, later on become popular as neo-classical school.
- The characteristic features of this new school was the introduction of marginal analysis.
- Jevons, Menger, and Walras were the founder of new school.
- These authors laid the foundation of demand or marginal utility ($MU \times P = DD$) of value.
- Alfred Marshall(1842-1924), one of the greatest figure in English economics, combine the classical's cost of production/supply side theory of value with the neo-classical demand side/marginal utility theory of value and formulated the famous MARSHALLIAN CROSS (The partial equilibrium.) ,i.e. demand and supply diagram and also he distinguishes short run and long run.

- Wicksell, Clark, Bawerk, Edgeworth, and Pareto further developed the the neoclassical theory of production and distribution.
- The neo-classical economics were concerned with the short run problem of allocation of given resources with maximum benefit.
- In neo-classical theory there is no possibility of economic crisis, because they assume market clearing hypothesis in the form of SAY'S LAW and full employment, thought there may be excess demand or excess supply in the market which is rectified by the relative price adjustment.
- Neo-classical believes what is true for an individual is true for economy as whole, i.e. aggregation of individual demand and supply schedule provide economy's demand and supply.—KEYNES criticize neo-classical economists on the ground of fallacy of composition

The Great Depression

- The Great Depression was a severe worldwide economic depression in the decade preceding World War II.
- It was the longest, most widespread, and deepest depression of the 20th century.
- The timing of the Great Depression varied across nations, but in most countries it started in 1930 and lasted until the late 1930s or middle 1940s.
- Unemployment in the U.S. rose to 25%, and in some countries rose as high as 33%. People wanted to work but could not find jobs at any wage.
- In United States Real GDP was 31% below its 1929 level.
- Personal income, tax revenue, profits and prices dropped, while international trade plunged by more than 50%.
- Some economies started to recover by the mid-1930s. In many countries, the negative effects of the Great Depression lasted until the

- Before the Depression, the prominent ideology was laissez-faire - keep the government out of the economy.
- Classical economics doesn't have any solution to the practical problem of Great Depression, except that to rely on market force to act and overtime in long run to balance relative price equilibrium.

John Maynard Keynes(1883-1946)

- Keynes was a British economist.
- His father, John Neville Keynes, was also an economist.
- A mathematician by training with special interest in subjective probability theory to model uncertain state of nature.
- Born and brought up at Cambridge in neo-classical tradition where Marshall and Pigou were among his teachers.
- Selected publications: Indian Currency and Finance(1913), The Economic Consequences of the Peace(1919), A Treatise on Probability(1921), The General Theory of Employment, Interest and Money(1936).

Keynes and the Great Depression

- The Great Depression was an intellectual failure for the economists working on business cycle theory—as macroeconomics was then called.
- In the 1930s, Keynes spearheaded a revolution in economic thinking, overturning the older ideas of neoclassical economics that held that free markets would, in the short to medium term, automatically provide full employment, as long as workers were flexible in their wage demands.
- Keynes instead argued that aggregate demand determined the overall level of economic activity, and that inadequate aggregate demand could lead to prolonged periods of high unemployment.
- According to Keynesian economics, state intervention was necessary to moderate “boom and bust” cycles of economic activity. He advocated the use of fiscal and monetary measures to mitigate the adverse effects recessions and depressions.

An Outline of Keynesian theory

- ① $\text{Employment} = \text{Output} = \text{Income}$. As employment increases, output and income increases proportionately.
- ② Volume of employment depends upon effective demand which in turn is determined by aggregate supply function (representing costs to entrepreneurs) and aggregate demand function (representing receipts of the entrepreneurs.) Keynes assumed aggregate supply function to be given in the short period.
- ③ Aggregate demand function is governed by consumption expenditure and investment expenditure.
- ④ Consumption expenditure depends upon size of income and propensity to consume. Consumption expenditure is fairly stable in the short period because propensity to consume does not change quickly.
- ⑤ Investment expenditure is governed by marginal efficiency of capital and the rate of interest. Unlike consumption expenditure, investment expenditure is highly unstable.

- ⑥ The marginal efficiency of capital is determined by the supply price of capital assets on the one hand and the prospective yield on the other. Prospective yield in turn depends upon future expectations. This explains why the marginal efficiency of capital and hence investment expenditure fluctuates.
- ⑦ Rate of interest is a monetary phenomenon and is determined by the demand of money (Liquidity Preference) and the quantity of money. Liquidity preference depends upon three motives:
 - a. Transaction motive
 - b. Precautionary motive
 - c. Speculative motive.

Investment function

- Because aggregate supply function is given and consumption expenditure is stable in short-run, the main reason for output fluctuation in short run is the fluctuations in highly unstable investment function.
- Private investment, according to Keynes, depends, on the one hand, MEC, i.e., the return on the capital, on the other hand, on rate of interest, i.e., the cost of investment.
- $MEC > ROI = \text{INVESTMENT INCREASES.}$
- $MEC < ROI = \text{INVESTMENT DECREASES.}$
- $MEC = ROI = \text{INVESTMENT} = 0.$

WHAT IS MEC?

- The MEC refers to the expected rate of return over cost from the employment of a brand new capital asset.

- In simple terms, the MEC = expected rate of profit of a capital asset.
- The MEC , in turn, depend upon a) prospective yield from the capital asset and b) the supply price of the capital asset.
- The supply price refers to the cost of producing a new capital asset. It is not the market price at which the capital asset is purchased, but is “the price which would just induce the manufacturer newly to produce an additional unit of such assets, i.e., what is sometimes called its replacement cost”.
- The prospective yield means the total returns which an entrepreneur expects to obtain from selling the output of the capital asset over its life time.
- Thus, the entrepreneur compares the prospective yield with the supply price in order to get an estimates of MEC .
- $MEC = SP =$ present value for future yields.

The neoclassical synthesis

- The neoclassical synthesis refers to a large consensus that emerged in the early 1950s, based on the ideas of Keynes and earlier economists.
- The theory was mainly developed by John Hicks, and popularized by the mathematical economist Paul Samuelson, who seems to have coined the term, and helped disseminate the “synthesis,” partly through his technical writing and in his influential textbook, *Economics*
- The IS–LM Model
- The most influential formalization of Keynes’s ideas was the IS-LM model, developed by John Hicks and Alvin Hansen in the 1930s and early 1940s.
- Theories of Consumption, Investment, and Money Demand
- In the 1950s, Franco Modigliani and Milton Friedman independently developed the theory of consumption, and insisted on the importance of expectations.

- James Tobin developed the theory of investment based on the relation between the present value of profits and investment. Dale Jorgenson further developed and tested the theory.
- William Baumol (1952) and James Tobin(1956) independently developed the model of the transactions demand for money. The theory relies on the tradeoff between the liquidity provided by holding money (the ability to carry out transactions) and the interest forgone by holding one's assets in the form of non-interest bearing money.
- Macro econometric Models
- Lawrence Klein developed the first U.S. macroeconomic model in the early 1950s. The model was an extended IS relation, with 16 equations.
- The neoclassical synthesis was to remain the dominant view for another 20 years. The period from the early 1940s to the early 1970s was called the golden age of macroeconomics.

The Breakdown of consensus

- The consensus in macroeconomics that prevailed until the early 1970s faltered for two reasons, one empirical and other is theoretical.
- The empirical reason is that the consensus view did not adequately cope with the rising rate of inflation and unemployment experienced during the 1970s.
- The theoretical reason is that the gap b/w microeconomic principles and macroeconomic practice was too great to be intellectually satisfying. (MANKIW NG, recent development in macroeconomics).

MONETARISM

- Monetarism is an empirical doctrine.
- Milton Friedman was the intellectual leader of the monetarists.
- In the 1960s, debates between Keynesians and monetarists dominated the economic headlines. The debates centered around three issues:
 - ① The effectiveness of monetary policy versus fiscal policy.
 - ② The Phillips curve.
 - ③ The role of policy.
- Monetary Policy versus Fiscal Policy

- Friedman in a theory of consumption function(1957) challenges the Keynes's notion of consumption as a function of current income, which provided the foundation for the fiscal policy multipliers that were central to the Keynesian theory and policy prescription. Instead he argued after empirically testing U.S. economy data that consumption is function of permanent income. If MPS out of transitory income is small, as Friedman's theory suggested, then fiscal policy would have a much smaller impact on equilibrium income than many Keynesian believed.
- Friedman challenged the view that fiscal policy could affect output faster and more reliably than monetary policy.
- In a 1963 book, A Monetary History of the United States, 1867-1960, Friedman and Anna Schwartz reviewed the history of monetary policy and concluded that monetary policy was not only very powerful, but that movements in money also explained most of the fluctuations in output.

- They interpreted the Great Depression as the result of major mistake in monetary policy.
- The Phillips Curve
- The Phillips curve provided a convenient way to complete Keynesian model, which always had trouble explaining why prices failed to equilibrate markets and how the price level adjusted overtime
- Friedman(1968) and Phelps(1967), took aim at the weakest link in the Keynesian model : the Phillips curve trade-off between inflation and unemployment.
- Phelps(1967) and Friedman(1968) argued that the trade-off b/w inflation and unemployment would not hold in long run, when classical principles should apply and money should be neutral.
- The trade-off appeared in the data because, in the short run, inflation is often unanticipated and unanticipated inflation can lower unemployment. The particular mechanism that Friedman suggest was money illusion on past of workers.

- By the mid 1970s, the consensus was that there was no long-run trade off between inflation and unemployment.
- The Role of Policy
- Skeptical that economists knew enough to stabilize output, and that policy makers could be trusted to do the right thing, Milton Friedman argued for the use of simple rules, such as steady money growth.
- Friedman believed that political pressures to “do something” in the face of relatively mild problems may do more harm than good.

New Classical Economics - The Rational Expectations Revolution

- Phelps(1967) and Friedman's(1968) natural rate hypothesis put expectation on the centre stage of Macroeconomic Literature.
- This prepared the way for the rational expectation revolution.
- In series of highly influential papers, Robert Lucas extended Friedman's argument.
- In his “Econometric Policy evaluation: A critique”, Lucas(1976) argued that the mainstream Keynesian Models were useless for policy analysis because they failed to take expectations seriously; as a result, the estimated empirical relationships that made up these models would break down if new policy were implemented.

- Lucas(1973) also proposed a business cycle theory based on the assumptions of imperfect information, rational expectations and market clearing.
- In this theory, monetary policy matters only to the extent to which it surprises people and confuses them about relative prices.
- Barrow(1977) offered evidence that this model is consistent with U.S. time series data.
- Sargent and Wallace(1975) pointed out a key policy implication: Because it is impossible to surprise rational people systematically, systematic monetary policy aimed at stabilizing the economy is doomed to failure.

New Classical Economics and Real Business Cycle Theory

- The 3rd wave of new classical economics was the real business theories of Kydland and Prescott(1982) and Long and Plosser(1983).
- Like the theories of Friedman and Lucas, these were built on the assumption that prices adjust instantly to clear the markets. But unlike the new classical predecessor, the real business cycle theories omitted any role of monetary policy, unanticipated or otherwise, in explaining economic fluctuations.
- The emphasis switched to the role of random shocks to technology and the intertemporal substitution in consumption and leisure that these shocks induced.
- Real business cycle (RBC) models assume that output is always at its natural level, and fluctuations are movements of the natural level of output. These movements are fundamentally caused by technological progress.

The New Keynesian

- New Keynesian economics is a school of contemporary macroeconomics that strives to provide microeconomic foundations for Keynesian economics. It developed partly as a response to criticisms of Keynesian macroeconomics by adherents of New Classical macroeconomics.
- Like the New Classical approach, New Keynesian macroeconomic analysis usually assume that households and firms have rational expectation. However they assume imperfect competition, with sticky prices and wages
- The new Keynesians are a loosely connected group of researchers working on the implications of several imperfections in different markets.

- One line of research focuses on the determination of wages in the labor market. George Akerlof has explored the role of “norms,” or rules that develop in any organization to assess what is fair or unfair.
- Another line of new Keynesian research has explored imperfections in credit markets. Ben Bernanke has studied the relation between banks and borrowers and its effects on monetary policy.
- Yet another direction of research is nominal rigidities in wages and prices. The menu cost explanation of output fluctuations, developed by Akerlof and N. Gregory Mankiw, attributes even small costs of changing prices to the infrequent and staggered price adjustment.

Recent development in Macroeconomics

- New neo-classical synthesis, is the fusion of the new classical economics and new Keynesian economics into a consensus on the best way to explain short-run fluctuations in the economy.
- New growth theory, New growth theory focuses on the determinants of technological progress in the long run, and the role of increasing returns to scale.
- This work drew attention away from short-run fluctuations, which had dominated the field of macroeconomics since its birth half a century earlier.
- 2008-09 Keynesian resurgence: In 2008 and 2009, there was a worldwide resurgence of interest in Keynesian economics among prominent economists and policy makers.

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